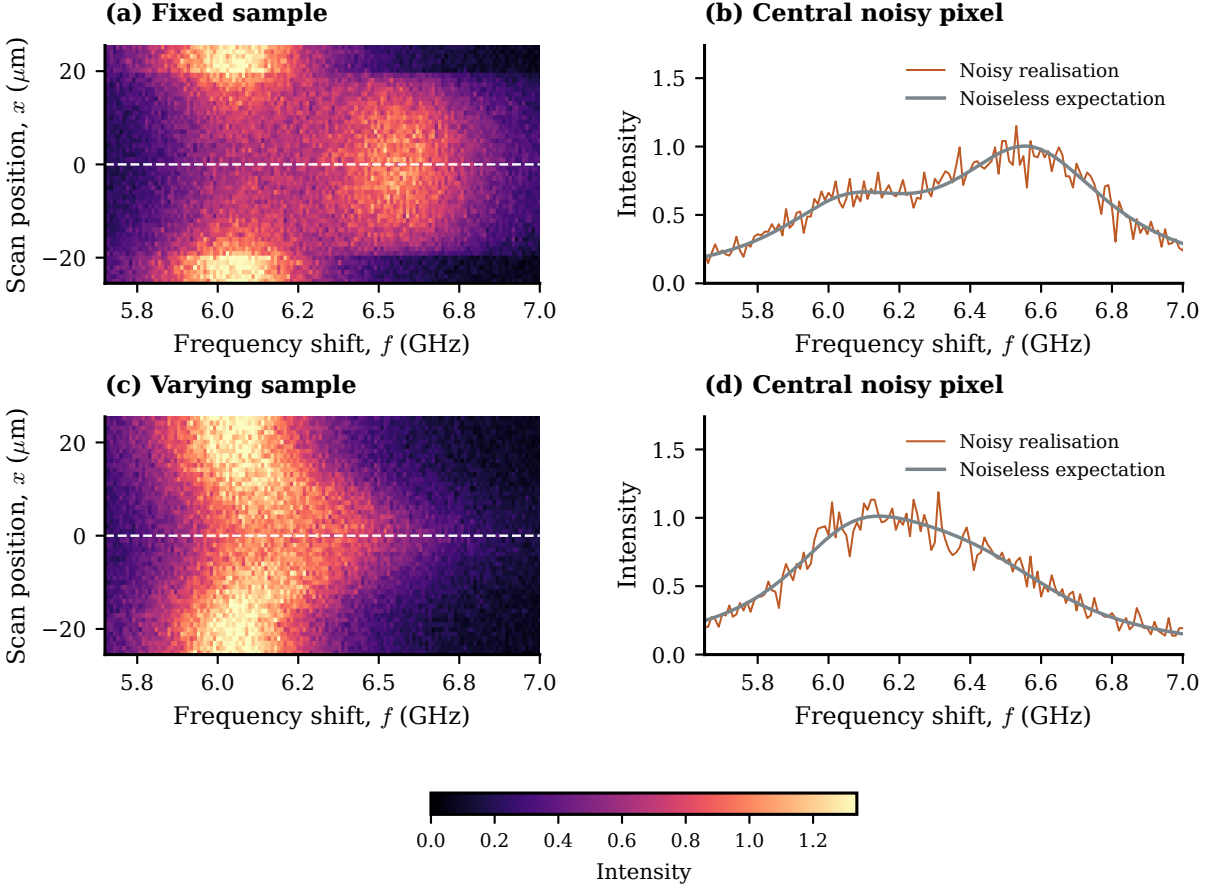

(a) Fixed sample
Scan position, x (μm)
20
0
−20
5.8
6.0
6.2
6.5
6.8
7.0
Frequency shift, f (GHz)
(b) Central noisy pixel
Noisy realisation
Noiseless expectation
Intensity
1.5
1.0
0.5
0.0
5.8
6.0
6.2
6.4
6.6
6.8
7.0
Frequency shift, f (GHz)
(c) Varying sample
Scan position, x (μm)
20
0
−20
5.8
6.0
6.2
6.5
6.8
7.0
Frequency shift, f (GHz)
(d) Central noisy pixel
Noisy realisation
Noiseless expectation
Intensity
1.5
1.0
0.5
0.0
5.8
6.0
6.2
6.4
6.6
6.8
7.0
Frequency shift, f (GHz)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Intensity